

Self Understanding Chain Rule

Name : _____

October 17

1. A composite function can be written $g(f(x))$ where g and f are basic functions. Identify the basic functions in the following.

(a) $h(x) = e^{-x^2}$, $g(x) = e^x$, $f(x) = -x^2$

(b) $h(x) = (x^2 + 3x - 2)^4$, $g(x) = x^4$, $f(x) = x^2 + 3x - 22$

(c) $h(x) = \sin(\log(2 * x + 2))$, $g(x) = \sin(x)$, $f(x) = \log(x)$, $j(x) = 2x + 2$

2. Find the derivative of the basic functions with respect to x in each of the above cases.

(a) $g(x) = e^x$, $g'(x) = e^x$, $f(x) = -x^2$, $f'(x) = -2x$

(b) $g(x) = x^4$, $g'(x) = 4x^3$, $f(x) = x^2 + 3x - 22$, $f'(x) = 2x + 3$

(c) $g(x) = \sin(x)$, $g'(x) = \cos(x)$, $f(x) = \log(x)$, $f'(x) = 1/x$ $j(x) = 2x + 2$, $j'(x) = 2$

3. Find the derivative of $h(x)$ with respect to x in each of the above cases.

4. $h(x) = e^{-x^2}$, $\frac{dh}{dx} = e^{-x^2} * (-2x)$

5. $h(x) = (x^2 + 3x - 2)^4$, $\frac{dh}{dx} = 4(x^2 + 3x - 2)^3 * (2x + 3)$

6. $h(x) = \sin(\log(2 * x + 2))$, $\frac{dh}{dx} = \cos(\log(2 * x + 2)) * \frac{1}{2*x+2} * 2$